

Federated Causal Diagnostics: Testing Assumption Quality Across Sites

Author: Daijiro Wachi (Independent OSS Engineer)

Code: <https://github.com/watilde/Harmonia>

Abstract

I built an open-source tool for diagnosing causal inference assumptions in federated settings. **One claim: diagnostic scores (0-1 scale) can be computed with $O(n)$ scaling and constant 150-byte communication, though exploratory thresholds (>0.8 for point estimation, 0.5-0.8 for bounds) lack empirical calibration.** Tested on 1,130-2,709,803 synthetic patients. Scores ranged 0.86-1.00 at 1k scale (CV=7.2%), converging at larger scales.

Note: This demonstrates computational feasibility using synthetic Synthea data. Threshold validation and real-world testing require institutional collaboration.

1. What I Built

A TypeScript tool that diagnoses three causal inference assumptions:

1. Unconfoundedness (no unmeasured confounding): - Standardized Mean Difference (SMD) - Propensity score overlap - Residual correlation

2. Positivity (common support): - Tail mass (extreme propensity scores) - Effective sample size

3. Specification (model fit): - R^2 (outcome model) - AUC (treatment model) - Calibration

Scoring: Each dimension $\rightarrow [0,1]$ score. Overall score = geometric mean.

Communication: Each site transmits 3 scores (24 bytes) + metadata = 50 bytes/site.

2. Test Results (Synthetic Data)

Table 1: Diagnostic Scores Across Three Scales

Scale	Patients	Site 1	Site 2	Site 3	Federated	CV	Processing
1k	1,130	0.95	0.86	1.00	0.95	7.2%	<1s

Scale	Patients	Site 1	Site 2	Site 3	Federated	CV	Processing
100k	235,222	0.97	0.96	0.97	0.97	0.5%	3s
2.8m	2,709,803	0.98	0.97	0.98	0.98	0.5%	13s

Key finding: All scores >0.8 (exploratory “safe” threshold), but threshold lacks calibration. Site heterogeneity (CV=7.2% at 1k) diminishes at scale (CV=0.5% at 2.8m).

Computational performance: - Overhead: ~30% added to baseline causal inference - Scaling: Linear $O(n)$ complexity - Memory: 2-3 GB per site

Communication: - Per site: 50 bytes (3 scores $\times$ 8 bytes + metadata) - Total: 150 bytes for 3 sites (constant across scales) - Reduction: 1k (1,340 $\times$), 100k (279,000 $\times$), 2.8m (3,200,000 $\times$)

3. Exploratory Thresholds (Unvalidated)

Proposed decision rules (require calibration):

Overall Score	Suggested Method	Interpretation
0.8	Point estimation	Assumptions likely satisfied
0.5 - 0.8	Partial ID bounds	Moderate concern
< 0.5	Sensitivity analysis	Severe violations

Critical caveat: These thresholds are **exploratory proposals**, not validated cutoffs. They require calibration via: 1. Controlled studies with known violations 2. Comparison to RCT ground truth 3. Empirical assessment of Type I/II error rates

Example (1k scale): Federated score = 0.95 suggests point estimation, but in real data this might be overconfident if diagnostics miss subtle violations.

4. How Scores Are Computed

Unconfoundedness score (0-1):

```
smd = max standardized mean difference across covariates
overlap = overlap coefficient of propensity distributions
residual_cor = correlation(outcome residuals, treatment residuals | covariates)

score = (1 - |smd|)  $\times$  overlap  $\times$  (1 - |residual_cor|)
```

Positivity score (0-1):

tail_mass = fraction with propensity <0.1 or >0.9
ess = effective sample size / n

score = (1 - tail_mass) × sqrt(ess)

Specification score (0-1):

r_squared = outcome model R^2
auc = treatment model AUC
calibration = Hosmer-Lemeshow p-value

score = (r_squared + auc + calibration_indicator) / 3

Overall score: Geometric mean of three dimensions.

5. Limitations

Threshold calibration: The 0.8 and 0.5 cutoffs are **not derived from empirical studies**. They are exploratory proposals requiring validation via controlled experiments with known assumption violations.

Synthetic data paradox: Synthea data has known causal structure, making it impossible to test diagnostic accuracy for real violations. We can verify *computational feasibility*, not *diagnostic validity*.

Diagnostic accuracy unknown: We don't know: - Sensitivity: Can diagnostics detect violations? - Specificity: Do they produce false alarms? - Calibration: Do scores correlate with bias magnitude?

Information leakage: Diagnostic scores reveal data quality indirectly. High scores suggest good covariate balance; low scores suggest treatment imbalance or poor model fit. This is partial privacy (better than sharing raw data, worse than zero disclosure).

Three sites only: Real networks have 10+ sites. Heterogeneity patterns may differ.

Purpose: This demonstrates **computational feasibility** (can we compute scores efficiently?), not **clinical validity** (should we trust these scores?).

6. Privacy Tradeoff

What is protected: Raw patient data and covariate identities (sites don't share individual records or variable lists).

What is disclosed: Diagnostic quality (scores indirectly reveal whether site has good balance, overlap, model fit).

Example: A score of 0.95 suggests strong covariate balance. A score of 0.60 suggests imbalance or model misspecification. Coordinators can infer data quality without seeing covariates.

Trade-off: This is inherent to diagnostic sharing. Hiding quality metrics would defeat the purpose of diagnostics.

7. Reproducibility

```
# Clone repository  
git clone https://github.com/watilde/Harmonia  
cd Harmonia
```

```
# Install dependencies  
npm install
```

```
# Build packages  
npm run build
```

```
# Run diagnostic tests  
npm run test:diagnostics:1k  
npm run test:diagnostics:100k  
npm run test:diagnostics:2.8m
```

Results saved to `research/modules/3-design-failure-aware-causal/experiments/results/`

8. What This Demonstrates

Computational feasibility: Diagnostics computed in seconds

Communication efficiency: 150 bytes (constant)

Heterogeneity detection: CV=7.2% at 1k, 0.5% at 2.8m

Integration: Works with existing causal inference pipeline

Threshold validation: 0.8/0.5 cutoffs lack calibration

Diagnostic accuracy: Unknown sensitivity/specificity

Real violations: Synthetic data can't test detection

Clinical deployment: Not ready without validation

9. Seeking Collaboration

I am an independent OSS engineer without access to real clinical data. This tool needs validation via:

1. **Controlled studies:** Inject known violations, test detection
2. **RCT comparison:** Compare observational diagnostics to RCT outcomes
3. **Multi-site testing:** Test on 10+ real hospital networks

If you have OMOP-formatted data, IRB approval, and expertise in causal inference, I welcome collaboration.

Contact: daijiro.wachi@gmail.com

References

1. Rosenbaum, P. R., & Rubin, D. B. (1983). The propensity score in observational studies. *Biometrika*, 70(1), 41-55.
2. Pearl, J. (2009). *Causality: Models, reasoning, and inference*. Cambridge University Press.
3. Stuart, E. A. (2010). Matching methods for causal inference. *Statistical Science*, 25(1), 1-21.
4. Petersen, M. L., et al. (2012). Diagnosing and responding to violations. *American Journal of Epidemiology*, 175(11), 1061-1071.
5. Austin, P. C. (2011). An introduction to propensity score methods. *Multivariate Behavioral Research*, 46(3), 399-424.